PRAVAN OMPRAKASH

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SUMMARY

Research engineer and computational materials scientist building first-principles, thermodynamic, and machine-learning workflows for materials discovery. First and corresponding author of a scalable phase-diagram prediction framework for refractory alloys and lead developer of RAPTor. Research spans rigorously benchmarked high-entropy-alloy prediction, reusable scientific software, theory-guided experimental collaboration, and DFT contributions to studies accepted in Science and Advanced Materials.

TECHNICAL SKILLS

Programming & scientific computing: Python, NumPy, Pandas, SciPy, Matplotlib, Plotly, Jupyter, Pymatgen
Machine learning: PyTorch, TensorFlow, scikit-learn, graph neural networks, CNNs, transfer learning, uncertainty quantification
Computational science: Density functional theory (VASP), molecular dynamics (LAMMPS), statistical thermodynamics, phase stability, inverse design

EXPERIENCE

PhD Candidate & Graduate Researcher | Washington University in St. Louis **Jan 2023-Present**
MCUBE Lab | St. Louis, MO

- First and corresponding author of Part I of a scalable first-principles framework that predicts 36 binary and 86 ternary refractory-alloy phase diagrams from DFT and sub-regular solution models; released on arXiv (2026).
- Benchmarked predictions against 51 experimental/CALPHAD phase diagrams, with most miscibility temperatures within +/-300 K; extended the workflow to 100+ experimental systems (81% accuracy) and 6,000+ higher-order alloy predictions.
- Engineered RAPTor's open-source Python workflows and public web interface for alloy phase stability, spinodal analysis, high-throughput screening, and interactive phase-diagram exploration.
- Co-designed SymPlex, a symmetry-conserving representation and visualization framework for high-dimensional alloy spaces; lead developer and co-first author in Scripta Materialia.
- Predicted hole doping lowers the coercive field of ferroelectric HfO2 from 8 to 6 MV/cm and reduces a switching barrier from 180 to 80 meV/f.u.; first-author study in Physical Review Materials (2026).
- Contributed DFT calculations to an experimental study of antiferroelectric hafnia accepted in Science; also co-authored an experimental-theory study of sub-nm2 charged domain-wall confinement in ZrO2 accepted in Advanced Materials.

Research Intern | Indian Institute of Science **Jun 2021-Mar 2022**
Bengaluru, India

- Developed graph and convolutional neural networks for material potential-energy surfaces used in molecular-dynamics simulations, including atomistic-data preparation, model evaluation, and uncertainty estimation.

Research Intern | IIT Bhubaneswar **Jan 2021-Feb 2022**
Bhubaneswar, India

- Built deep-learning surrogate models for inverse design of mechanical metamaterials and optimization of piezoelectric vibration-energy harvesters; translated finite-element simulation data into fast design-space exploration tools.

SELECTED MACHINE-LEARNING SYSTEMS

- Perovskite property prediction - curated a cross-chemistry crystal dataset and trained graph-representation models for band-gap prediction (0.28 eV MAE; millisecond inference); first author in Computational Materials Science.
- AuthNet - co-developed a language-independent, contactless facial-authentication system using identity and temporal facial movements (98.1% test accuracy); AAAI 2021 Top 20 Student Abstract.

EDUCATION

PhD, Materials Science and Engineering | Washington University in St. Louis **Expected Dec 2026**
St. Louis, MO | GPA: 3.75/4.00

BTech, Metallurgical and Materials Engineering | National Institute of Technology Karnataka **2018-2022**
Surathkal, India | GPA: 8.6/10

SELECTED PUBLICATIONS & RECOGNITION

- Omprakash, P. et al. “First-principles-based Prediction of Phase Fields: Part I. Binary and Ternary Refractory Alloys.” arXiv:2609.06627 (2026). First and corresponding author.
- Omprakash, P. et al. “Hole-doping reduces the coercive field in ferroelectric hafnia.” Physical Review Materials 10, 034416 (2026).
- Cavin, J.*, Omprakash, P.* et al. “SymPlex Plots for Visualizing Properties in High-Dimensional Alloy Spaces.” Scripta Materialia (2025). *Equal contribution; MRS Fall 2025 Best Oral Presentation.
- Li, X. et al. “Antiferroelectric Hafnia Down to the 2D Limit.” Science (accepted). Contributing author; DFT calculations.
- Afroze, N. et al. “Sub-nm2 ferroelectric domains via charged 180 degree walls in ZrO2.” Advanced Materials (accepted). Contributing author.